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**Female-founded
semiconductor AI startup
SixSense raises \$8.5M**



A Singapore-based deep tech startup called [SixSense](#) has developed an AI-powered platform that helps semiconductor manufacturers predict and detect potential chip defects on production lines in real time.

It has raised \$8.5 million in Series A bringing its total funding to around \$12 million. The round was led by Peak XV's Surge (formerly Sequoia India & SEA), with participation from Alpha Intelligence Capital, FEBE, and others.

Founded in 2018 by engineers Akanksha Jagwani (CEO) and Avni Agrawal (CTO), SixSense aims to address a fundamental challenge in semiconductor manufacturing: converting raw production data, from defect images to equipment signals, into real-time insights that help factories prevent quality issues and improve yield.

Despite the sheer volume of data generated on the fab floor, what stood out to the co-founders was a surprising lack of real-time intelligence.

Akanksha brings a deep understanding of manufacturing, quality control, and software automation through her experience building automation solutions for manufacturers like Hyundai Motors and GE and led product development at startups like Embibe. Agarwal adds technical experience from her time at Visa, where she built large-scale data analytics systems, some of which were later protected as trade secrets. A

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skilled coder with a strong background in mathematics, she had long been interested in applying AI to traditional industries beyond fintech.



IMAGE CREDITS:SIXSENSE

inspection processes remain largely manual and fragmented, Agarwal told TechCrunch. After speaking with more than 50 engineers, it became clear there's significant room to modernize how quality checks are done, she added.

Fabs today are filled with dashboards, SPC charts, and inline inspection systems, but most only display data without further analysis, Agarwal said. "The burden of using it for decision-making still falls on engineers: [they must] spot patterns, investigate

Together, the duo evaluated sectors from aviation to automotive before landing on semiconductors. Despite the semiconductor industry's reputation for precision,

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anomalies, and trace root causes. That's time-consuming, subjective, and doesn't scale well with increasing process complexity."



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SixSense provides engineers with early warnings to address potential issues before they escalate with capabilities such as defect detection, root cause analysis, and failure prediction.

SixSense's platform is also specifically designed to be used by process engineers rather than data scientists, Agarwal said. "Process engineers can fine-tune models using their own fab data, deploy them in under two days, and trust the results — all without writing a single line of code. That's what makes the platform both powerful and practical."

The competitive landscape includes in-house engineering teams using tools like Cognex and Halcon, inspection equipment makers integrating AI into their systems, and startups including Landing.ai and Robovision.

SixSense’s AI platform is already in use at major semiconductor manufacturers like GlobalFoundries and JCET, with more than 100 million chips processed to date. Customers have reported up to 30% faster production cycles, a 1-2% boost in yield, and a 90% reduction in manual inspection work, the founders said. The system is compatible with inspection equipment that covers over 60% of the global market.

“Our target customers are large-scale chipmakers — including foundries, outsourced semiconductor assembly and test providers (OSATs), and integrated device manufacturers (IDMs),” Agarwal said. “We’re already working with fabs in Singapore, Malaysia, Taiwan, and Israel, and are now expanding into the U.S.”

Geopolitical tensions, especially between the U.S. and China, are reshaping where chips are made, driving new manufacturing investments across the globe.

“We’re seeing fabs and OSATs expand aggressively in Malaysia, Singapore, Vietnam, India, and the U.S. — and that’s a tailwind for us. Why? Because we’re already based in the region, and many of these new facilities are starting fresh — without legacy systems weighing them down. That makes them far more open to AI-native approaches like ours from day one,” Agarwal told TechCrunch.

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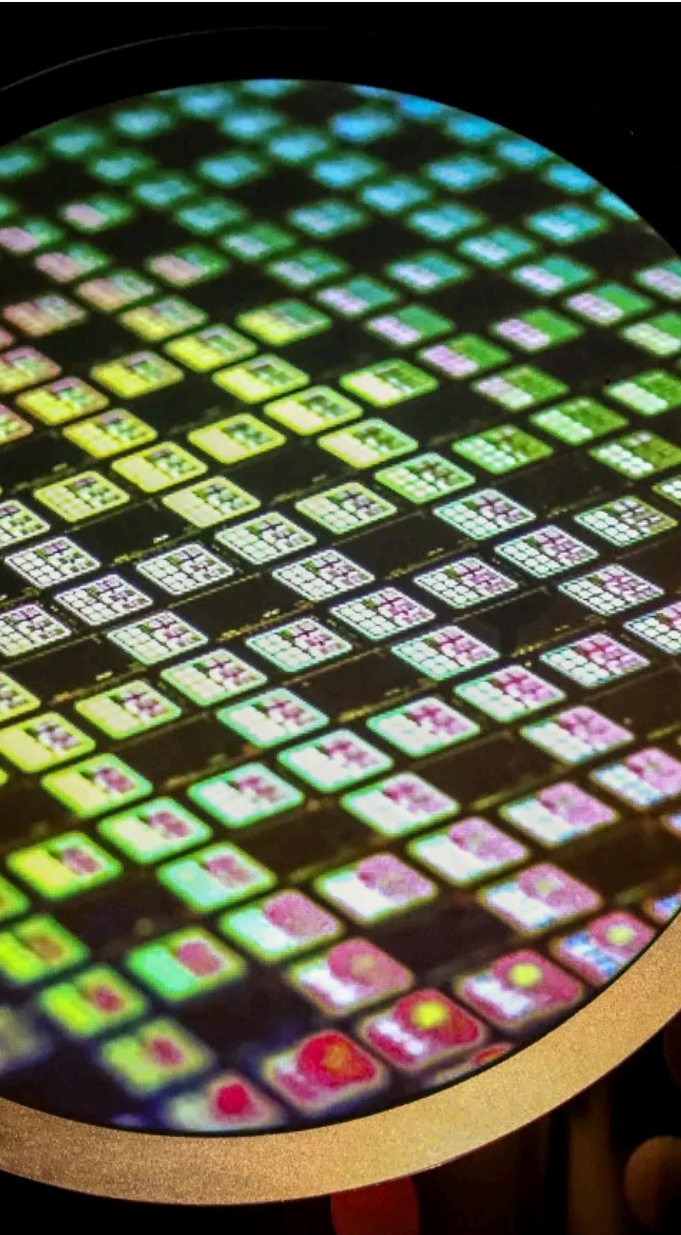
Kate Park

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US to award TSMC \$6.6B in grants, \$5B in loans to step up chip manufacturing in Arizona

Kate Park — 6:07 AM PDT · April 8, 2024

IMAGE CREDITS:

I-HWA CHENG / BLOOMBERG / GETTY IMAGES

The U.S. Commerce Department [said](#) on Monday that it has signed an agreement to award Taiwan Semiconductor Manufacturing (TSMC) \$6.6 billion in direct funding under the CHIPS and Science Act to

set up semiconductor factories in Phoenix, Arizona, and provide up to \$5 billion in loans.

This grant, pegged for the company's U.S. subsidiary, TSMC Arizona, is the latest step by the U.S. to strengthen its domestic supply of semiconductors as it seeks to reshore manufacturing of chips amid escalating geopolitical tensions between the U.S. and China.

The [CHIPS Act](#), signed into law in 2022, earmarks an investment of about \$280 billion to boost domestic chip research and production in the U.S, of which about \$52 billion has been set aside to subsidize domestic chip manufacturing. Besides national security concerns arising from semiconductors primarily being made in Asia, a big motivator for the U.S. is to diversify the production of semiconductors, and bring more electronics production to the West. The Act is primarily aimed at attracting manufacturing stateside, and also prohibits recipients of the funding from increasing their semiconductor manufacturing footprint in China.

With the new investment, Taiwan-based TSMC, which is the world's largest producer of semiconductors, is broadening its plans for its fabrication plants in Arizona. The company said it would build a third fabrication unit in addition to the two being built right now, and will manufacture 2-nanometer or more advanced chips. The company had previously said it would invest [about \\$40 billion](#) to set up plants in the U.S.

TSMC said its first fab unit is slated to begin producing chips under the 4nm process in the first half of 2025; the second factory will produce 3nm and 2nm chips from [2028](#); and the third plant will start manufacturing 2nm and more advanced chips near the end of the decade.

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TSMC is investing more than \$65 billion via these projects in the U.S., and the company said in a statement that the investment makes this the largest ever direct investment by a foreign entity in a greenfield project in the U.S.

TSMC Arizona will sell its chips to its U.S. customers, which include AMD, Apple, Nvidia and Qualcomm. The company expects the three fab units to create approximately 6,000 direct high-tech, high-wage jobs, and more than 20,000 construction jobs.



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The White House last month said it had signed an agreement with the Department of Commerce [to grant Intel up to \\$8.5 billion](#) to shore up U.S.-based production.

Intel [could receive approximately \\$20 billion in grants and loans](#) from the CHIPS and Science Act for its semiconductor manufacturing. Meanwhile, Samsung, [which announced a \\$17 billion additional investment](#) in Taylor, Texas, [is expected to receive](#)

[more than \\$6 billion in grants](#) for its chip facility in Texas.

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Samsung commits
\$230B for five new chip
plants in South Korea

Samsung Electronics said today that it plans to invest approximately \$230 billion (300 trillion won) to build five new memory and foundry fabs in South Korea — a big move in line with the government's ambitious aim to set up a mega semiconductor hub in Yeosu, on the outskirts of Seoul. The investments will be made through 2042.

The company's move indicates that it is shoring up its domestic semiconductor production line to secure the supply chain as other countries, including the U.S., Taiwan, Japan and China, are scrambling to ramp up their domestic chip manufacturing to offset risks to global supply chain disruption due to rising tensions between the U.S. and China.

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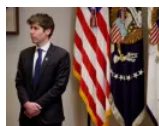
"It is expected that we would invest about 300 trillion KRW (\$230 billion) in the chip-making cluster through 2042," a spokesperson at Samsung said in an emailed statement to TechCrunch. Although the government, in a statement, spoke of plans for five plants, the Samsung spokesperson declined to comment on the number of plants Samsung will set up in the semiconductor cluster, as well as other details.

South Korea's Ministry of Trade, Industry and Energy (MOTIE) [unveiled](#) Wednesday its new project to invest \$422 billion (500 trillion won) by 2026 to promote six core technologies: semiconductors, electric vehicle batteries, autonomous vehicles, robots, displays and biotechnology. The government [said](#) it would earmark \$260 billion (340 trillion won)

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specifically for the chip space to develop system semiconductors by 2026, considering “semiconductors as strategic economic support and national security assets.”

The mega semiconductor hub also will have a whole value chain of chip-making processes from memory, foundry and design houses to material suppliers and attract 150 domestic and global fabless companies and advanced chip materials and equipment makers, per the announcement by the country’s trade ministry. The South Korean government wants to foster high-tech industries by working with corporations and intends to offer expanded tax breaks for companies in the advanced tech space, the statement says.

South Korea is not the only country to be making big moves to build out its own manufacturing operations.

Japan has been partnering with global semiconductor and chip equipment makers to revive its own semiconductor industry. The world’s largest contract chip producer, Taiwan Semiconductor Company (TSMC), also has been expanding its manufacturing footprint both domestically and overseas in the U.S. and Japan.



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[SoftBank, NEC, Sony, Toyota + more team up for Rapidus, Japan's bid for next-gen chip domination](#)

Samsung already operates a foundry chip facility in Austin, Texas, and it has recently announced additional investment plans for the U.S.: \$17 billion earmarked to build a manufacturing facility in Taylor, Texas. In addition, it is considering [investing \\$200 billion to set up a further 11 chip plants in Texas](#).

Separately, Korea's Ministry of Science and ICT announced last month to boost its AI chip industry with \$642.5 million.

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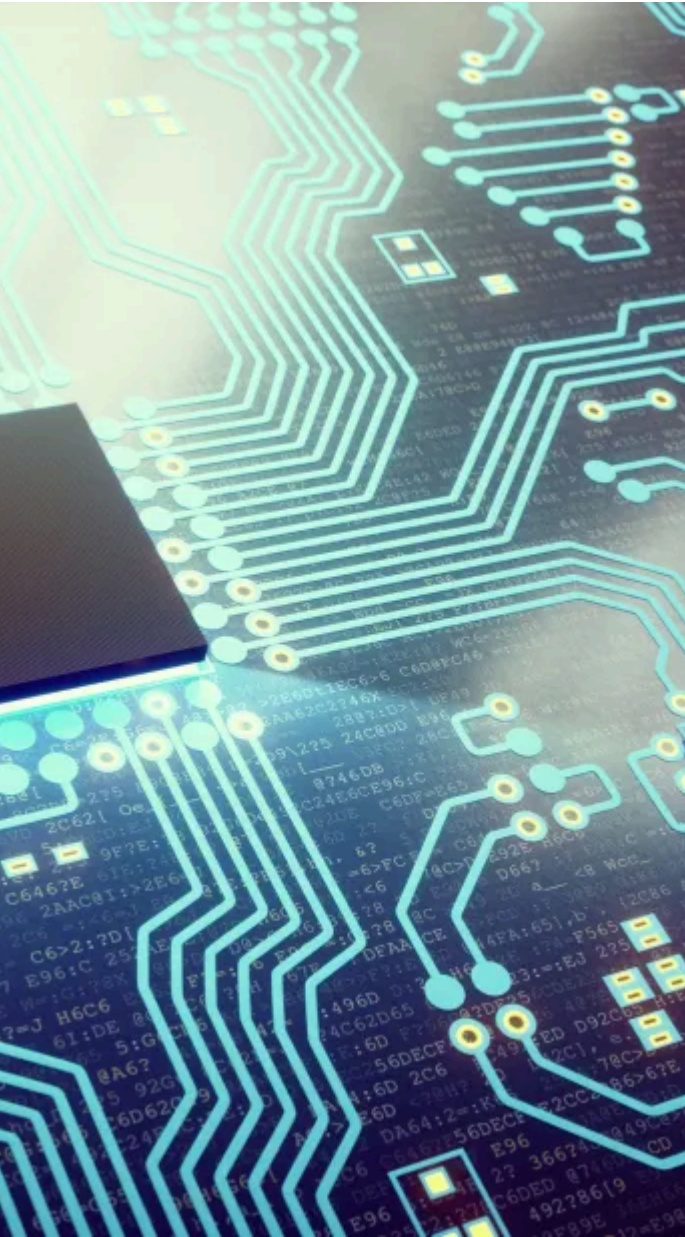


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For the CHIPS Act to pay off, the US needs a historic investment in memory production

Brian Shirley — 9:00 AM PDT · October 13, 2023

In the year since President Biden signed the CHIPS and Science Act, there has been significant attention paid to this landmark legislation to rebuild the U.S’ semiconductor industry. We’ve seen numerous announcements of historic private-sector investments focused on strengthening domestic semiconductor manufacturing, supply

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chain and R&D. Yet most of the national conversation has focused on logic chips.

There is another category of semiconductors that is at risk of being forgotten: memory chips. Since the invention of the first commercially viable “dynamic random-access memory” (DRAM) in the early 1970s, memory has been one of the key enablers of modern computing systems, helping drive the overall semiconductor industry. Virtually all modern electronics, including cell phones, cars, medical gear, military systems, and cloud data centers, are dependent on large amounts of memory to safely store the information on which we rely. And given the insatiable demands of today’s aptly named “data economy,” this dependence grows stronger every year: Memory is mission-critical to the technologies of the future, such as AI and supercomputing.

Memory technologies are at the heart of enabling and sustaining our nation’s innovation competitive edge, yet [less than 2% of the world’s memory supply is manufactured within the U.S.](#), and all of that comes from a single company, Micron Technology. Currently, every leading-edge memory fab is [located in Asia](#), with more being built every year as nations realize the importance of memory. This has formed an overconcentration of manufacturing capacity and supply chain risk that’s significantly more dangerous than that found in the logic industry.

Domestic production of secure memory technology is critical to U.S. technology leadership, jobs and economic growth, as well as our national and economic security, and thus requires immediate attention.

As someone who built a career in memory and serves as an adviser to companies and the government, I can vouch that memory is a brutally

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competitive business. Last year, memory represented about [28% of the global \\$556 billion](#) semiconductor market, about the same percentage as logic. However, while often overshadowed by the larger advertising and branding budgets of logic companies, the workhorse memory industry drives a [far larger share](#) of the industry's capital spending.

New memory fabs must be built [significantly larger](#) than most logic fabs to be competitive. Keeping up with the most advanced technology drives [massive R&D bills](#), as well as steep ongoing capital reinvestment. And these investments must be made every year, despite profitability that is far lower on average and much more volatile than found in the logic industry.

Unfortunately, U.S. companies aren't competing on a level playing field. The memory industry owes its start to U.S. innovation, but today is concentrated in Asia where governments have responded to the industry's economics by subsidizing memory production and R&D.



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Securing the supply chain for these essential system ingredients should be reason enough for the U.S. to invest in memory. The integrity of every computing system we rely on is directly dependent on devices that store our critical information. Geopolitical disruption of the memory supply chain would be catastrophic to every segment of the economy, but making an investment in memory has additional tremendous benefits for the U.S.

First, memory drives the lion's share of the demand for production semiconductor equipment. While the U.S. still leads this industry, the benefit goes to offshore nations intent on monopolizing memory. A U.S. memory investment would help ensure this equipment enables our own production first, in addition to providing the large domestic volume demand for everything from raw materials to silicon wafers to the chemicals and gasses on which semiconductor production relies.

Second, the size and scale of memory fabs would allow the U.S. to truly revitalize the domestic skills base in fab construction and operation, providing a [stable base of well-paying jobs](#) that cannot rely on logic fab construction alone for sustainability.

Finally, increased investment in new memory fabs would be [large enough to economically transform entire communities](#) and regions. And the stimulus benefits of the core investment are multiplied by the economic activity generated in local communities and along the supply chain.

We are at a critical juncture, and the Biden administration must ensure that some of the largest opportunities created by the CHIPS and Science Act are leveraged on behalf of the memory industry. Semiconductor memory is foundational to our nation's technical leadership and yet is more at risk than ever. As policymakers debate the merits of different semiconductor projects, the economic

and national security benefits of a historic investment in leading-edge memory production are sure to pay off for the U.S. We cannot forget about memory.

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Brian Shirley

Contributor

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